Elyssa Hofgard

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EDUCATION

2022–present **PhD Electrical Engineering and Computer Science (EECS)**
Massachusetts Institute of Technology, expected graduation June 2027 GPA: 4.90/5.00

2021–2022 **M.S. Computational and Mathematical Engineering**
Stanford University GPA: 3.97/4.00

2017–2021 **B.S. Physics with Honors, Minors: Mathematics and History**
Stanford University GPA: 3.91/4.00

RESEARCH INTERESTS

- Electronic structure, topology, and correlations in disordered and amorphous quantum materials
- Approximate symmetry and symmetry-based descriptors of local and medium-range order
- Emergent order and multipolar order parameters in amorphous and magnetic systems
- Symmetry-preserving and equivariant machine learning methods for materials characterization and discovery

RESEARCH & EMPLOYMENT EXPERIENCE

2022–present **Massachusetts Institute of Technology**
Graduate Research Assistant, Atomic Architects, Professor Tess Smidt
- Developing E(3)-equivariant neural network architectures for applications in materials science and physics
- Incorporating interpretable symmetry breaking parameters into E(3)-equivariant networks to model phase transitions
- Determining crystal structure and lattice parameters from powder X-ray diffraction data using invariant, symmetry-aware representations

2024–present **Lawrence Berkeley National Laboratory**
DOE CSGF Practicum, Dr. Sinéad Griffin & Dr. Dmitriy Morozov
- Two summer practicums through the DOE CSGF (2024, 2026), continued collaboration
- Characterizing length scales of order in amorphous materials using persistent homology, modeling multipolar order parameters in magnetic materials using ML and DFT codes
- Developed an efficient implementation of structural spillage calculations to identify band inversion in non-crystalline topological materials

2021–2022 **Tompkins Laboratory**
Research Assistant, IRIS-HEP
- Funded as a research assistant through Institute for Research and Innovation in Software for High Energy Physics (IRIS-HEP)
- Project focused on improving Kalman Filter algorithms used in particle track reconstruction

Summer 2021 **Lawrence Livermore National Laboratory**
Data Science Summer Institute (DSSI)
- Project focused on improving seasonal predictions of precipitation and temperature over the western US using deep learning

2020–2021 **ATLAS Experiment, CERN**
Research Assistant, Professor Lauren Tompkins
- 1 of 11 students selected nationwide for the International Research Experience for Students (IRES) at CERN funded through the National Science Foundation
- Honors thesis explored the use of machine learning techniques to search for dark photons in ATLAS data, improved analysis sensitivity 10 times
- Thesis awarded Firestone Medal for Excellence in Undergraduate Research

2019–2021 **Molecular Imaging Instrumentation Laboratory**
Research Assistant, Professor Craig Levin
- Developed a computational framework for line-of-response identification in simultaneous multi-isotope PET

Summer 2019 **Regeneron Genetics Center**
Summer Intern, R&D Data Engineering Group
- Developed a statistical method to distinguish gene level effects across phenotypes

2018–2019 **LIGO Group**
Research Assistant, Professor Brian Lantz
- Designed a wind fence for seismic isolation at the LIGO Hanford Site

HONORS & AWARDS

2024 **QuARC, MIT QSEC Annual Conference**
Best Poster in Quantum Algorithms and Machine Learning Session.

2023 **Department of Energy Computational Science Graduate Fellowship**
Awarded, up to 4 years of funding

2023 **National Defense Science and Engineering Graduate Fellowship**
Declined for DOE CSGF

2022 **MIT Advanced Television and Signal Processing Fellowship**

2022 **NSF GRFP Honorable Mention**

2021 **Firestone Medal for Excellence in Undergraduate Research**
Awarded for thesis with ATLAS experiment
- Given to the top ten percent of all honors theses in the social sciences, natural sciences, and engineering and applied sciences at Stanford

2019 **American Physical Society Far West Section Meeting**
Honorable Mention for Best Undergraduate Poster

PUBLICATIONS

[6] **E. Hofgard**, K. Min, N. Segal, D.W. Mittan-Moreau, A. Mansouri Tehrani, V. Oklejas, J. Nigam, D.W. Paley, A.S. Brewster, T. Smidt, "Learning Lattice Parameters from Powder X-Ray Diffraction Data Using Invariants," *Acta Crystallographica Section A: Foundations and Advances*, 2026. Accepted.

[5] H. Lawrence*, **E. Hofgard***, V. Portilheiro, Y. Chen, T. Smidt, R. Walters, "To Augment or Not to Augment? Diagnosing Distributional Symmetry Breaking," *International Conference on Learning Representations*, 2026. *denotes equal contribution.

[4] **E. Hofgard**, A. Mansouri Tehrani, Y. Lin, S. Ji, T. Smidt, "Materials Characterization" (Section 7.4, pp 581–586), in "Artificial Intelligence for Science in Quantum, Atomistic, and Continuum Systems," *Foundations and Trends in Machine Learning*, Now Publishers, 2025, Vol. 18: No. 4, pp 385–912. Link: `https://doi.org/10.1561/2200000115`.

[3] R. Wang, **E. Hofgard**, H. Gao, R. Walters, and T. Smidt, "Discovering Symmetry Breaking in Physical Systems with Relaxed Group Convolution," *Proceedings of the Forty-first International Conference on Machine Learning*, PMLR 235, July 2024, pp. 50599–50621. Link: `https://proceedings.mlr.press/v235/wang24y.html`.

[2] The ATLAS collaboration, "Search for dark photons from Higgs boson decays via ZH production with a photon plus missing transverse momentum signature from pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector," *J. High Energ. Phys.*, Springer, 2023, article 133. Link: `https://doi.org/10.1007/JHEP07(2023)133`. *Work recognized by the 2025 Breakthrough Prize in Fundamental Physics.*

[1] D.J. Cotter, **E. Hofgard**, J. Novembre, Z.A. Szpiech, and N.A. Rosenberg, "A Rarefaction Approach for Measuring Population Differences in Rare and Common Variation," *Genetics*, Oxford University Press, 224(2), article iyad070, pp. 1–8 (2023). Link: `https://doi.org/10.1093/genetics/iyad070`.

COLLABORATION PUBLICATIONS

As a member of the LIGO Scientific Collaboration (2018–2019), co-author on 11 LSC publications on gravitational wave observations and detector characterization; full list available via Google Scholar.

PREPRINTS

[3] R. McConkey, J. Balla, **E. Hofgard**, T. Smidt, A. Bodner, "Rotational equivariance and locality in data-driven subgrid-scale closures," arXiv:2607.26850 (2026).

[2] R. McConkey, J. Balla, J. Bailey, A. Backour, **E. Hofgard**, T. Jaakkola, A. Bodner, T. Smidt, "Turbulence teaches equivariance to neural networks," arXiv:2602.04695 (2026).

[1] R.B. Garg, **E. Hofgard**, L. Tompkins, H. Gray, "Exploration of Different Parameter Optimization Algorithms within the Context of ACTS Software Framework," arXiv:2211.00764 (2022).

WORKSHOP PAPERS

[3] J. Balla*, J. Bailey*, A. Backour, **E. Hofgard**, T. Jaakkola, T. Smidt†, R. McConkey†, "Implicit Augmentation from Distributional Symmetry in Turbulence Super-Resolution," *Machine Learning and the Physical Sciences Workshop, NeurIPS* (2025). Link: `https://arxiv.org/abs/2509.20683`. *, †Equal contribution.

[2] **E. Hofgard**, R. Wang, R. Walters, T. Smidt, "Relaxed Equivariant Graph Neural Networks," *GRaM Workshop at the 41st International Conference on Machine Learning* (2024). Link: `https://arxiv.org/abs/2407.20471`.

[1] K. Min, **E. Hofgard**, T. Smidt, "FrOGS: Discrete Neural Sampler for Independent Alloy Configurations Across Chemical Conditions," arXiv:2609.02948 (2026). NeurIPS 2026 AI4Mat Workshop, Spotlight presentation. Undergraduate mentee (MIT UROP) as first author.

CONFERENCES & TALKS

INVITED TALKS

[2] **E. Hofgard**. "Using E(3)-Equivariant Neural Networks (ENNs) to Uncover Symmetry-Implied Missing Information." Computational Research in Boston and Beyond (CRIBB) Seminar, March 2025.

[1] **E. Hofgard**. "Applications of Euclidean Neural Networks for the Design and Understanding of Physical Systems." Plenary talk at The 5th International Conference on Data-Driven Plasma Science, Berkeley, CA. Given on behalf of Tess Smidt, August 2024.

CONFERENCE PRESENTATIONS AND POSTERS

[8] **E. Hofgard**, G. Moore, O. Ashour, E. Banyas, T. Smidt, S.M. Griffin. "Using Equivariant Neural Networks to Learn Multipolar Order Parameters." Oral Presentation at APS Global Physics Summit 2026.

[7] **E. Hofgard**, D. Morozov, M.A. Hussien, T. Bayaraa, S.M. Griffin, T.E. Smidt. "Applications of Topological Data Analysis (TDA) for Studying Structural Motifs in Amorphous Bi_2Se_3." Oral Presentation at APS Global Physics Summit 2025.

[6] **E. Hofgard**, A.M. Tehrani, and T. Smidt. "Learning Crystal Structure from Powder X-Ray Diffraction Data Using Invariants." Oral Presentation at 2024 American Physical Society March Meeting.

[5] **E. Hofgard**, R. Wang, and T. Smidt. "Using Symmetry-Equivariant Neural Networks (ENNs) to Uncover Symmetry-Implied Missing Information." Oral Presentation at 2023 Materials Research Society Fall Meeting.

[4] **E. Hofgard** et al. "Exploration of Machine Learning (ML) Methods for Determining Unit Cell Parameters from Powder XRD Patterns." Oral Presentation at 2023 Materials Research Society Fall Meeting.

[3] **E. Hofgard** and L. Tompkins. "Cuts Optimization and Machine Learning Models for Dark Photon Signal-Background Discrimination with the ATLAS Detector." Oral presentation at 2021 APS April Meeting.

[2] **E. Hofgard**, G. Chinn, and C. Levin. "Simultaneous Multi-Isotope PET: A Computational Framework for Line of Response (LOR) Identification." Presented at 2020 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC). Link: `https://doi.org/10.1109/NSS/MIC42677.2020.9507891`.

[1] **E. Hofgard**. "An Algorithmic Approach to Identify Non-Pure Positron Emitters for Positron Emission Tomography (PET) Imaging Applications." Poster presented at 2019 Annual Meeting of the APS Far West Section.

SOFTWARE

Structural Spillage Package: Developer of a more efficient implementation of structural spillage calculations for identifying band inversion in non-crystalline topological materials, extending Muñoz-Segovia *et al.*, *Phys. Rev. Research* 5, L042011. `https://github.com/Griffin-Group/Structural-Spillage-Package`.

TEACHING & ADVISING

Fall 2026 — **Teaching Assistant, MIT 18.C06 Linear Algebra and Optimization**

- Responsibilities include leading twice-weekly recitations for 40 students and holding office hours in a 180-student course

2025, 2026 — **Guest Lecturer, MIT 6.7970/8.750: Symmetry and its Application to Machine Learning (Symm4ML)**

- Guest lectures on symmetry breaking and relaxed equivariance in Prof. Tess Smidt's graduate course

2024–present — **Mentor, MIT Undergraduate Research Opportunities Program**

- Supervising Kyucheol Min, an undergraduate MIT student, on representation learning for XRD data, resulting in co-authorship on Hofgard, Min, et al. "Learning Lattice Parameters from Powder X-Ray Diffraction Data Using Invariants."
- Min's first-author work, "FrOGS: Discrete Neural Sampler for Independent Alloy Configurations Across Chemical Conditions," was selected for a spotlight presentation at the NeurIPS AI4Mat Workshop (2026)

Fall 2020 — **Teaching Assistant, Stanford Physics 21: Mechanics, Fluids, and Heat**

Winter 2020 **Learning Assistant, Stanford Physics 41E: Mechanics**

2019–2021 **Tutor, Schwab Learning Center, Stanford University**

- Provided individual tutoring for students with learning differences; tutored linear algebra, multivariable calculus, and differential equations

PROFESSIONAL DEVELOPMENT

2026 **Selected Participant, MIT Path of Professorship Workshop**

- Competitive workshop preparing female-identifying graduate students and postdocs for academic careers

2025 **Participant, Los Alamos Computational Condensed Matter Summer School**

- Topics covered include density functional theory, the many-body problem, experimental spectroscopy

PROFESSIONAL SERVICE

REVIEWING

- International Conference on Learning Representations (ICLR), 2026
- Conference on Neural Information Processing Systems (NeurIPS), 2026
- Geometry-grounded Representation Learning and Generative Modeling (GRaM) Workshop at ICML/ICLR, 2024, 2026
- Machine Learning and the Physical Sciences (ML4PS) Workshop at NeurIPS, 2024-2025
- Acta Crystallographica Section A, 2026

LEADERSHIP

2024–2025 **Committee Member, IAIFI Equity and Inclusion Committee**

NSF AI Institute for Artificial Intelligence and Fundamental Interactions

- One of two graduate student members on the committee to advise IAIFI on supporting early career members

2023–2024 **Co-President of Graduate Women in Course 6 (GW6)**

- Elected as co-president of GW6, an organization supporting MIT EECS graduate women

2022–2023 **MIT EECS Graduate Student Council (GSC) Representative**

- Elected to advocate for the EECS student body

2020–2021 **Undergraduate Representative, Physics Undergraduate Studies Committee**

- Selected as one of two undergraduate representatives to offer ideas on course design and department culture

2018–2020 **Leadership, Physics Undergraduate Women and Gender Minorities at Stanford**

- Co-President, 2019–2020; Director of Community Development, 2018–2019

OUTREACH

2026 **Invited Panelist, MIT EECS New PhD Cohort Orientation**

- Orientation event welcoming the incoming 2026 PhD cohort (113 students)

2025–present **Mentor, MIT EECS Buddy Program**

- Mentoring and advising first-year PhD students on the transition to the PhD

2024–2025 **Invited Representative, MIT EECS Recruitment & Outreach**

- Panelist, MIT EECS Graduate Admissions Outreach, sharing the graduate student experience with prospective PhD students (2025)
- Selected by the EECS department to represent MIT at the Society of Women Engineers (SWE) Conference (2024)

2024 **Featured Researcher, Foundry Fireside Chats, Molecular Foundry, LBNL**

- Public video series explaining research at the Molecular Foundry to a general audience
- Interview titled "From Data to Discovery: Machine Learning's Insight into the World of Topological Amorphous Materials" (video link)